

考研精读 | Economist



Well Informed

心里有数——健康专栏

难度评级: ★★★☆☆

Can you **hack** your **gut microbiome**?

你能黑进自己的肠道微生物组吗？

【 Para.1 】

The trillions of **bugs** in a human gut digest complex **carbohydrates** into fatty acids that provide about 10% of an individual's calories. They **synthesise** vitamins, notably K (which regulates blood **clotting**) and various members of the B group. They help **tune** the immune system. They **suppress** the growth of **pathogens**. And they regulate production of signalling molecules, such as **serotonin**, involved in sending messages from the gut to the brain.

人体肠道中数以万亿计的微生物会将复杂碳水化合物消化分解成脂肪酸，为人体提供约10%的热量。它们还能合成维生素，尤其是维生素K（能够调节血液凝固）以及多种B族维生素；帮助调节免疫系统；抑制病原体的生长；并调控血清素等信号分子的生成，向大脑传递肠道信息也有这些信号分子的功劳。

August 15TH- 21ST | Science & technology

文章导读：

文章指出肠道微生物承担消化、合成维生素、调节免疫等重要功能，多样饮食与发酵食品可提升菌群多样性。不少商家推出粪便检测及定制化肠道调理服务，但并无标准健康菌群作参照，相关论断科学依据薄弱，检测也存在偏差，调理效果仅微乎其微，专家建议不如选择普通饮食调整。

hack v. /hæk/ to find a way of looking at and/or changing information on sb else's computer system without permission 非法侵入（他人的计算机系统）；黑客攻击【[考研大纲词汇](#)】

The bank's security system was **hacked** by a group of cybercriminals. 该银行的安全系统遭到一群网络罪犯的非法侵入。

gut n. /gʌt/ the tube in the body through which food passes when it leaves the stomach 消化道；肠道【[考研大纲词汇](#)】

guts n. /gʌts/ (informal) the courage and determination that it takes to do sth difficult or unpleasant 勇气；胆量；决心；毅力

microbiome n. /maɪkrəʊˈbaɪəʊm/ the community of microorganisms (such as bacteria, fungi, and viruses) that live in a particular environment, especially in the human body 微生物组；微生物群落

bug n. /bʌg/ (informal) an illness that is usually not serious and is caused by bacteria or a virus （引起小毛病的）细菌；病毒【[考研大纲词汇](#)】

carbohydrate n. /ˌkɑːbəʊˈhaɪdreɪt/ a substance such as sugar or starch that consists of carbon, hydrogen and oxygen. Carbohydrates in food provide the body with energy and heat. 碳水化合物；糖类【[考研大纲词汇](#)】

synthesise v. /ˈsɪn θ əsaɪz/ (also synthesize) to produce a substance by a chemical reaction in plants or animals 合成【[考研大纲词汇](#)】

There are many vitamins that the body cannot **synthesize** itself. 有很多维生素是人体自身无法合成的。

ANT break down

clot v. /klɒt/ when blood or cream clots or when sth clots it, it forms thick, almost solid masses （血液或奶油）凝结成块；使凝结【[考研大纲词汇](#)】

He was rushed into hospital because his blood wasn't **clotting** properly. 他被匆忙送进医院，因为他的血液不能正常凝结。

n. an almost solid piece of something 凝块

He had a blood **clot** removed from his brain. 他颅内的一个血块被取了出来。

n. a stupid person 傻瓜，笨蛋

Look what you've done, you **clot**! 看看你干了什么，笨蛋！

【 Para.2 】

Keeping your gut microbiome in **tip-top** condition is thus wise. Maintaining a diverse mix of bacteria in it is widely thought a good way to do this. And, since different species have different **appetites**, diversity of diet makes sense.

因此，保持肠道微生物组处于最佳状态实属明智之举。人们普遍认为，维持肠道内菌群多样化有助于保证微生物群的良好状态。而且，由于不同菌种的口味各不相同，饮食种类丰富也十分重要。

【 Para.3 】

In 2018 an American citizen-science project showed that people eating 30 types of plants a week had significantly more diverse microbiomes than those eating fewer than ten. And work from Stanford University, published in 2021, confirmed that **fermented** foods such as kimchi and sauerkraut diversify microbiomes, too. Some people, though, feel such **off-the-peg** advice is **insufficient**. They would prefer something **bespoke**.

2018年，一项由美国公众参与的科学研究项目显示，每周食用30种植物食物的人，其肠道微生物组多样性明显高于每周食用不足10种的人。斯坦福大学2021年发表的一项研究也证实，泡菜、酸菜等发酵食品同样能够增加肠道微生物组的多样性。不过，有些人觉得这种一刀切式的建议还不够。他们更希望得到量身定制的方案。

tune v. /tju:n/ to prepare or change sth so that it is suitable for a particular situation 调整；使适合；使协调【**考研英语大纲词汇**】

His speech was **tuned** to what the audience wanted to hear. 他的演讲根据听众的期望进行了调整。

be in/out of tune with someone/something idiom if you are in tune with people or ideas, you understand or agree with them, and if you are out of tune with them, you do not. 与...协调 / 不协调，与...相 / 不相契合

Much of his success comes from being in **tune with** what his customers want. 他的成功很大程度上归功于他契合了顾客的需要。

suppress v. /sə'pres/ to prevent something from being seen or expressed or from operating 抑制，忍住，压抑；查禁【**考研英语大纲词汇**】

The virus **suppresses** the body's immune system. 这种病毒会抑制人体的免疫系统。

后缀：**press**按，压

compress v. /kəm'pres/ (被)压紧 (com- 共同，一起 + press 按，压)

depress v. /di'pres/ 使沮丧 (de- 离开，向下 + press 按，压)

oppress v. /ə'pres/ 压迫 (op- 逆，相反 + press 按，压)

pathogen n. /'pæ θ ədʒən/ any small organism, such as a virus or a bacterium that can cause disease 病原体【**考研英语大纲词汇**】

serotonin n. /serə'təʊnɪn/ a chemical in the brain that affects how messages are sent from the brain to the body, and also affects how a person feels 血清素【**考研英语大纲词汇**】

tip-top adj. /'tip.tɒp/ excellent; perfect 非常优秀的，完美的
Even though our house is in **tip-top** condition, we're having problems selling it. 虽然我们的房子状况很好，但现在把它卖出去还是有困难。

appetite n. /'æpɪtaɪt/ the feeling that you want to eat food 食欲，胃口【**考研英语大纲词汇**】

n. a strong desire for something 强烈欲望；渴望

Birds have different **appetites** for different types of insects. 不同鸟类对不同类型的昆虫有着不同的偏好。

ferment v. /fə'ment/ if food or drink ferments or if you ferment it, it goes through a chemical change because of the action of yeast or bacteria, which may cause it to produce bubbles or heat, or turn sugars in it into alcohol. (使) 发酵【**考研英语大纲词汇**】

You make wine by leaving grape juice to **ferment** until all the sugar has turned to alcohol. 酿葡萄酒是让葡萄汁发酵，直到所有的糖分都转化为酒精。

off-the-peg adj. /,ɒf.ðə'peg/ (US off-the-rack); (Australian English off-the-hook) made and bought in standard sizes and not made especially to fit a particular person (衣服) 均码的，标准码的；非定制的

insufficient adj. /,ɪnsə'fɪʃnt/ not enough 不够的，不足的；不充分的【**考研英语大纲词汇**】

bespoke adj. /br' spəʊk/ specially made for a particular person 定做的；定制的

fistful n. /'fɪst.fʊl/ an amount of something that you can hold in your fist 一把 (之量)

【 Para.4 】

For a suitable **fistful** of dollars, they can have it. **So-called** direct-to-consumer (D2C) firms will analyse **stool** samples and **compile** lists of the bugs **therein**, based on the DNA and RNA they find. Such firms may also offer **tailored** dietary recommendations, doses of live bacteria and special bug-friendly **supplements**, all intended to fill any microbial blanks that their tests have **flagged up**.

只要花上一笔钱，他们就能得到这样的服务。所谓直接面向消费者（D2C）的公司会分析粪便样本，根据其中检测到的DNA和RNA，列出样本中所含微生物的清单。这类公司可能还会提供个性化的饮食建议、活菌剂量以及专门有利于微生物生长的补充剂，旨在填补检测所发现的微生物缺失。

【 Para.5 】

The problem with this approach is that there is no "right" microbiome to act as a reference. In fact, microbiomes can differ quite a bit from each other while still being perfectly effective at delivering a desirable outcome to their host. The biggest red flag for poor gut health is a simple lack of biodiversity—which is what a varied and partly fermented diet is good at correcting. By contrast, the idea that **tinkering** with the presence or absence of this or that type of bug will perceptibly improve your health has **scant** published scientific basis, and many D2C firms are reluctant to share the details of **proprietary** data on which they base their claims.

这种方法的问题在于，可以作为参照标准的所谓“正

so-called adj. /səʊ 'kɔ:ld/ used to show that you think a word that is used to describe sb/sth is not suitable or not correct 所谓的

adj. used to introduce a new word or phrase that is not yet known by many people 叫做...的，名叫...的；所谓的

stool n. /stu:l/ a piece of solid waste from the body 粪便

compile v. /kəm 'paɪl/ to collect information from different places and arrange it in a book, report, or list 汇编；编纂；编辑；编制【考研英语词汇】

We're **compiling** some facts and figures for a documentary on the subject. 我们正在为一部有关这个主题的纪录片汇编资料。

therein adv. /ðeər 'ɪn/ in or into a particular place, thing, etc. 在那里；在其中；在那点上

It is a thrilling tale of a haunted house and the ghosts **therein**. 那是一个令人毛骨悚然的故事，讲的是一座凶宅还有里面的幽灵。

tailored adj. /'teɪləd/ (of clothes) made to fit a particular person by a tailor (衣服) 量身定制的，合身的

supplement n. /'sʌplɪmənt/ something that is added to something else in order to improve it or complete it; something extra 补充物，增补物【考研英语词汇】

v. /'sʌp.l.ment/ to add something to something to make it larger or better 增加，补充，增补

flag up phrasal verb to draw attention to sth 引起对...的注意
The report **flagged up** the dangers of under-age drinking. 这篇报道引起大众对未成年人饮酒危害的关注。

tinker v. /'tɪŋkə(r)/ to make small changes to something, especially in an attempt to repair or improve it 修补；改进；摆弄

He spent the afternoon **tinkering** with his bicycle chain. 他花了一下午摆弄他的自行车链条。

scant adj. /skænt/ very little; not enough 很少的；不足的；欠缺的

adj. almost 接近；差不多

proprietary adj. /prə 'praɪətri/ (especially of information, software, or a product) made and sold by a particular company and protected by a patent, trademark, or copyright from being copied or used by others (尤指信息、软件或产品) 专有的，专卖的，专利的

commerce n. /'kɒmɜ:s/ the activity of buying and selling goods and services, especially in large amounts; trade 商业；贸易；（尤指大宗商品的）买卖【考研英语词汇】

The new online platform has revolutionized cross-border **commerce**. 这个新的在线平台彻底改变了跨境贸易。

respectable adj. /rɪ 'spektəbl/ considered by society to be acceptable, good, or of sufficient quality; deserving respect and approval 体面的；值得尊敬的；相当好的；可接受的【考研英语词汇】

The study was published in a **respectable** scientific journal. 该研究发表在一本相当权威的科学期刊上。

respectful adj. /rɪ 'spektfəl/ showing admiration for someone or something 毕恭毕敬的；表示尊敬的；尊重的

"We're so pleased to meet you at last," he said in a **respectful** tone of voice. “我们终于和您见面了，真是万分高兴，”他毕恭毕敬地说道。

确”微生物组并不存在。事实上，不同人体内的微生物组可能存在相当大的差异，却依然能够非常有效地为宿主带来理想的结果。肠道健康不佳最明显的警示，是生物多样性不足，而多样化、且部分经过发酵的饮食恰恰有助于纠正这一问题。相比之下，认为通过调整某一种微生物的存在与否，就能让健康状况得到明显改善，这一观点几乎没有什么已发表的科学依据；而且，许多D2C公司也不愿公开其相关主张所依据的专有数据的具体细节。

【 Para.6 】

There are exceptions. In Britain, Tim Spector has ridden the twin horses of academic science (a post at King's College, London) and **commerce** (he helped found a D2C firm called Zoe) without so far falling off. Results from Zoe's customers are fed, with their permission, into **respectable**, peer-reviewed research. So far, this suggests individually tailored diets do not have huge effects, but may help at the **margins**.

当然，也有例外。在英国，蒂姆·斯佩克特（Tim Spector）横跨学商两界，他在伦敦国王学院任职的同时，还参与创办了一家名为Zoe的D2C公司。到目前为止，他在两个身份之间来回切换，游刃有余。经客户许可，Zoe公司收集到的用户数据会被纳入正规的同行评议研究。截至目前，这些研究表明，个性化饮食并不会产生特别显著的效果，但或许能在一些细微之处有所帮助。

microbial adj. /maɪ'krəʊbiəl/ relating to or caused by microbes (= very small living things, especially bacteria, that can cause disease) 微生物的；由微生物引起的
The research focuses on the **microbial** communities living in the human digestive system. 这项研究聚焦于生活在人类消化系统中的微生物群落。

census n. /'sensəs/ a count for official purposes, especially one to count the number of people living in a country and to collect information about them (官方的) 调查，统计；(尤指) 人口普查 【考研大纲词汇】

accurate adj. /'ækjərət/ correct and true in every detail 准确的；精确的；正确无误的 【考研大纲词汇】
The scientist needed **accurate** measurements to verify the experiment results. 这位科学家需要精确的测量数据来验证实验结果。

homogenise v. /hə'mɒdʒənaɪz/ to treat milk so that the cream is mixed into other parts of the liquid 使均质（尤指使牛奶中的油脂粒均匀分布）
The dairy industry **homogenizes** milk to prevent the cream from rising. 乳制品行业将牛奶均质化，以防止油脂漂浮上升。
v. to change something so that all its parts or features become the same or very similar (使) 变均匀；(使) 变纯一；(使) 变类同
TV has **homogenized** the culture and language of large parts of the planet. 电视使地球上大部分地区的文化和语言同质化。

identical adj. /aɪ'dentɪkl/ exactly the same; not different in any way 完全相同的；一模一样的 【考研大纲词汇】
The two samples produced almost **identical** results in the laboratory test. 这两个样本在实验室测试中得出了几乎完全相同的结果。

put v. /pʊt/ to express or state something in a particular way 说；表达；表述
To **put** it simply, the experiment failed because of human error. 简单地说，实验失败是由于人为失误。

bogus adj. /'bəʊgəs/ (especially of a claim or a document) not genuine; pretending to be real or true (尤指说法或文件) 假的；伪造的；冒充的 【考研大纲词汇】
The insurance company suspected the claim was **bogus** and launched an investigation. 该保险公司怀疑这项索赔是伪造的，并展开了调查。

【 Para.7 】

However, even if a bespoke approach does have value, it must surely depend on the **microbial censuses** carried out by D2C firms being **accurate**. To test that, researchers at America's National Institute of Standards and Technology prepared 21 sets of **homogenised** (and thus **identical**) faeces and sent three to each of seven D2C firms. When the results (which were published in February) came back there were notable differences not only between different firms' analyses but also, in one case, between those of an individual company.

不过，即便量身定制的方法确实有价值，它显然也必须建立在D2C公司所进行的微生物普查足够准确的基础上。为了检验这一点，美国国家标准与技术研究院的研究人员制备了21份经过均质处理、因而完全相同的粪便样本，并给向7家D2C公司每家寄出了3份。检测结果（已于2月发表）寄回后，人们发现，不仅不同公司的分析结果之间存在明显差异，甚至有一家公司的检测结果内部也有不同。

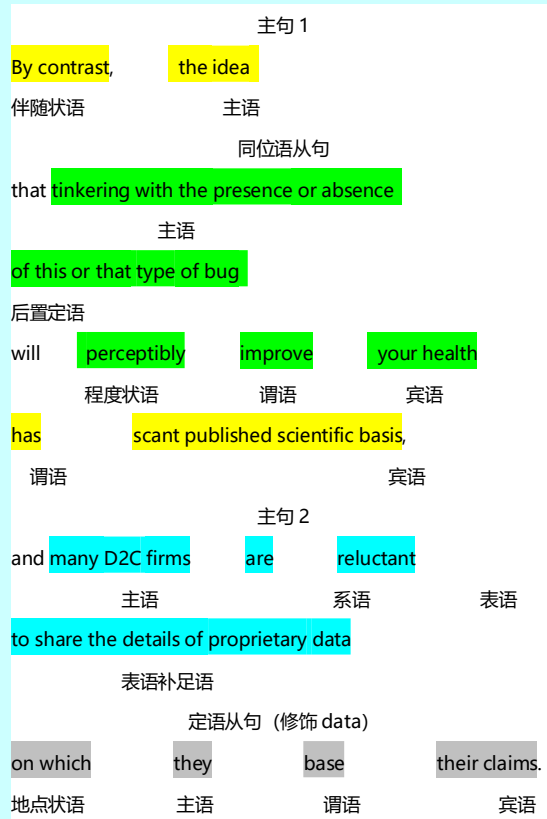
【 Para.8 】

As Jonathan Eisen, an evolutionary microbiologist at the University of California, Davis, observes, just because microbiomes differ does not mean an era of personalised microbiome medicine is coming any time soon. In fact, he **puts** it more strongly than this, saying, "Most of the claims [in this area] are **bogus**." Perhaps better to save your money and invest in a jar of kimchi.

加州大学戴维斯分校的进化微生物学家乔纳森·艾森

►今日句子分析

Para. 5 By contrast, the idea that tinkering with the presence or absence of this or that type of bug will perceptibly improve your health has scant published scientific basis, and many D2C firms are reluctant to share the details of proprietary data on which they base their claims. 相比之下，认为通过调整某一种微生物的存在与否，就能让健康状况得到明显改善，这一观点几乎没有什么已发表的科学依据；而且，许多D2C公司也不愿公开其相关主张所依据的专有数据的具体细节。



句子分析：

主句1：

伴随状语：By contrast；（by contrast对比前文某种比较有科学依据、看起来具有健康改善作用的观点/证据。而本句开始转向：作者认为，通过调整微生物来改善健康缺乏科学依据。注意，however也可以用于对比，区别在于however表示转折，而by contrast表示“相比之下”，更强调比较两个对象之间的差异。）主语：the idea；

同位语从句：（解释idea的内容。从句是主谓宾结构，已经是一个完整的句子，that没有在里面充当任何成分，所以它不是定语从句。）

引导词：that；主语：tinkering with the presence or absence；（tinker with 不是简单的change，而是对某物进行小幅调整、改动，带有“试着改改看”“随意调整”的意味。）

后置定语：of this or that type of bug；（this or that在这里指的不是前文已有内容，不表示“这个或那个”，而是表示“某一种或另一种”，简言之就是“某种”。如You can choose this or that option. 你可以选择某个选项。

the presence or absence（存在或不存在）of this or that type

(Jonathan Eisen) 指出，微生物组存在差异，并不意味着个性化微生物组医学时代就会很快到来。其实，他本人的话更为直接：“（这个领域的）大多数说法都是骗人的。”所以，你也许还是省下这笔钱，买罐泡菜更好。

➤ 写作积累

1. (Para. 6) So far, this suggests individually tailored diets do not have huge effects, but may help at the margins.

at the margins 表示 “being at the outer edge, border, or boundary of a group, place, or system, often used to describes people, ideas, or activities that have little power, influence, or involvement in the main part of society”，即“只在有限程度上、在细微之处，仅带来边际性的”，例如，Individually tailored diets may make a difference at the margin, even if their overall impact is limited. 即使个性化饮食的总体影响有限，它在某些细微方面仍可能产生一定作用。此外，还有以下表达可以表示类似的含义：

yield modest gains idiom to produce small, limited, or average improvements, profits, or increases rather than large or dramatic breakthroughs 获得轻微的收益，收效不大

The stock market recovery will yield modest gains for investors this year. 今年的股市复苏只会为投资者带来适度收益。

move the needle idiom to make a noticeable, measurable difference or progress toward a goal 带来可感知的变化（与 at the margins 并不是完全同义，而是反向强调），产生足以被注意到的变化

The treatment barely moves the needle. 这种治疗几乎没有产生什么实质性改变。

2. (Para. 7) However, even if a bespoke approach does have value, it must surely depend on the microbial censuses carried out by D2C firms being accurate.

Carry out 表示 “to perform, complete, or execute a task, plan, order, or promise”，即“实施、开展、进行（某项工作、研究、调查、检测等）”，例如，It is hard to carry out the new plan in a short time. 在短时间内落实这个新计划很难。此外，还有以下表达可以表示类似的含义：

put something in place idiom to set up, establish, or implement a system, rule, plan, or physical object so it is ready to use 实施，使就位

The government put a new tax policy in place this year. 政府

of bug (某一种微生物的)，字面意思是“某一种微生物的存在或不存在”，太啰嗦。译文简化为“某一种微生物的存在与否”。)

程度状语：perceptibly；谓语：will improve；宾语：your health；谓语：has；宾语：scant published scientific basis；主句2：

连词：and；主语：many D2C firms；系语：are；表语：reluctant；（and并列两个分句，分句1：the idea ... has scant published scientific basis，分句2：many D2C firms are reluctant to share...。这句话的结构就是[A has B], and [C are reluctant to D].)

表语补足语：to share the details of proprietary data；

定语从句（修饰data）：

地点状语：on which；主语：they；谓语：base；宾语：their claims；（their指many D2C firms）

今年出台了一项新的税收政策。

go through with something idiom to do a planned or promised action, especially when it is hard, scary, or you feel unsure about it 把...真正做下去，执行到底

We have decided to go through with the original plan despite the high cost. 尽管代价高昂，我们还是决定按原计划把事情进行到底。